

level 1

tow truck

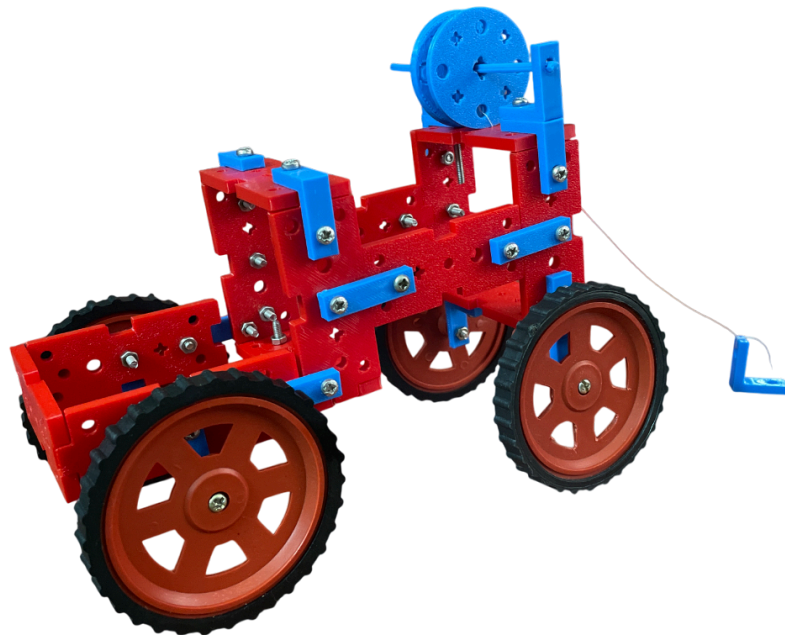
I have a hook and a pulley system to pull heavy cars. When you turn my handle, the string winds up and lifts the  weight of the car – showing how simple machines like pulleys and wheels do the heavy work!"

no. of sessions: 6

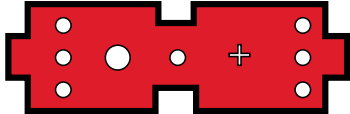
simple machine:

wheel, axle, pulley

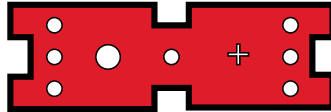
oh.



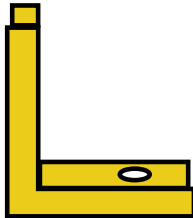
materials needed



m-m beam - x4



f-f beam - x4



L clamp - x8

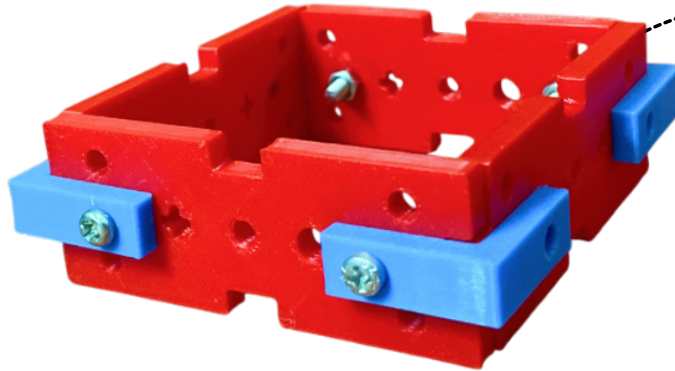


small screw - x16



nut - x16

step 1: assemble the back chassis



Let us start by building the chassis of the Tow Truck.

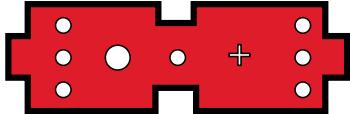
Join **2 x F-F Beams** and **2 x M-M Beams** together to form a square.

Use **4 x L Clamps** to hold them together.



Repeat step 1 to make another set.

materials needed



m-m beam - x1



small screw - x2

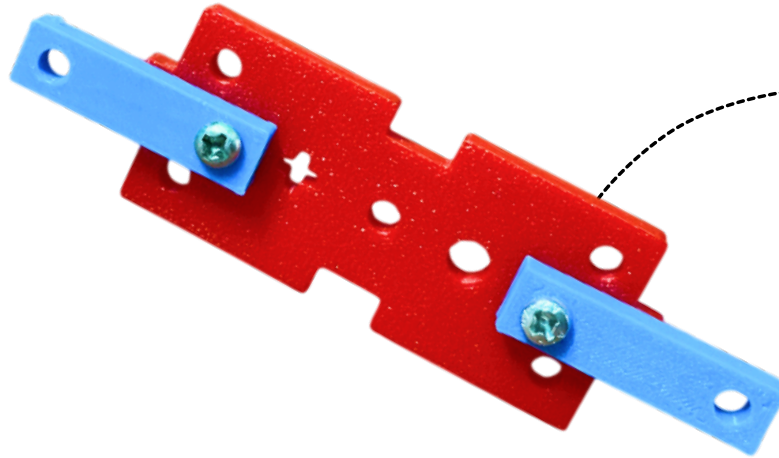


nut - x2



I clamp - x2

step 2: assemble the chassis supports



Join **2x 1 clamps** to the ends of the **M-M beams** as shown.



Repeat the step to make another set.

materials needed

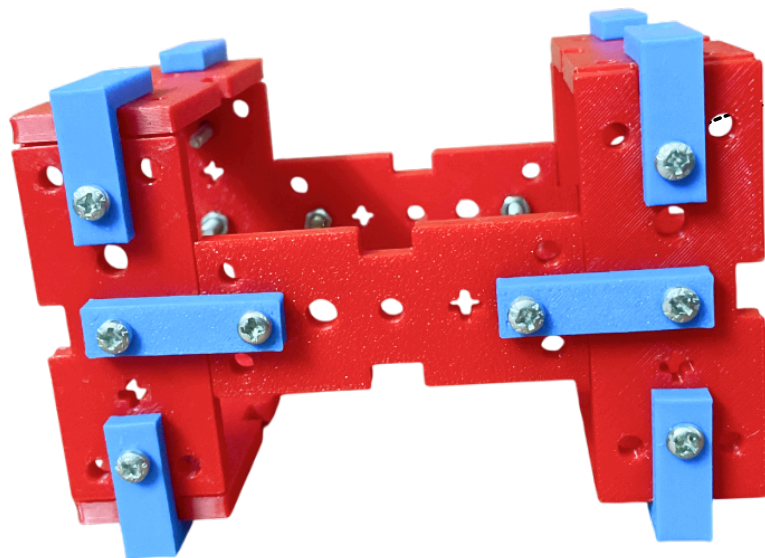


small screw - x4



nut - x4

step 3: complete the back chassis



Join the **2 square pieces** together using the supports as shown.

materials needed



I clamp - x2

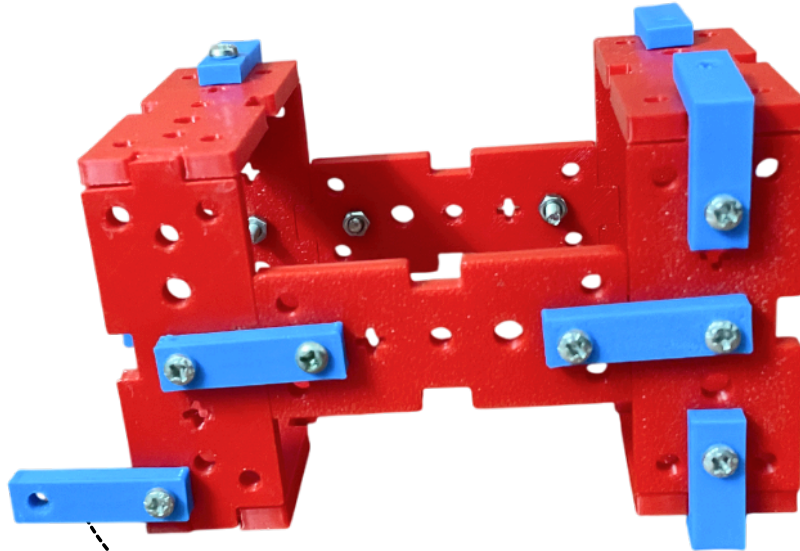


small screw - x2



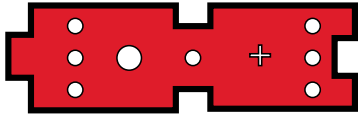
nut - x2

step 4: assemble the front chassis supports

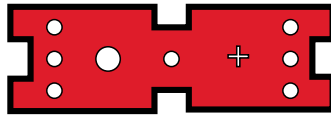


Attach **2x 1** clamps
to both sides of the
chassis as shown.

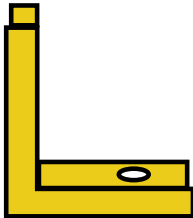
materials needed



m-f beam - x2



f-f beam - x1



L clamp - x2

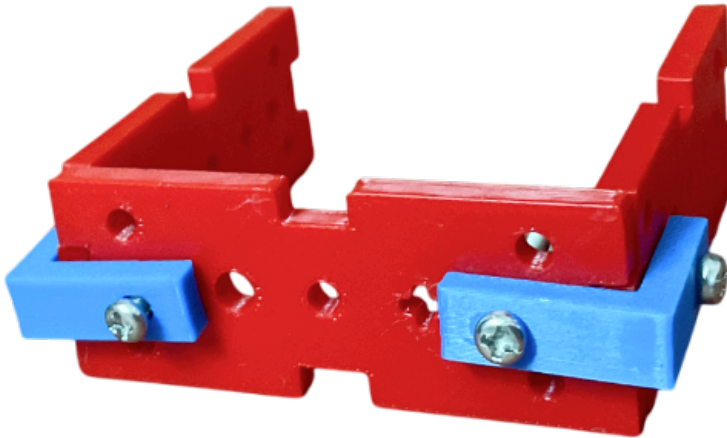


small screw - x4



nut - x4

step 5: assemble the front chassis supports



Lets join **2 x M-F** and **1 F-F beam** to form a U shape.

Use **2 L clamps** to hold them together.

materials needed

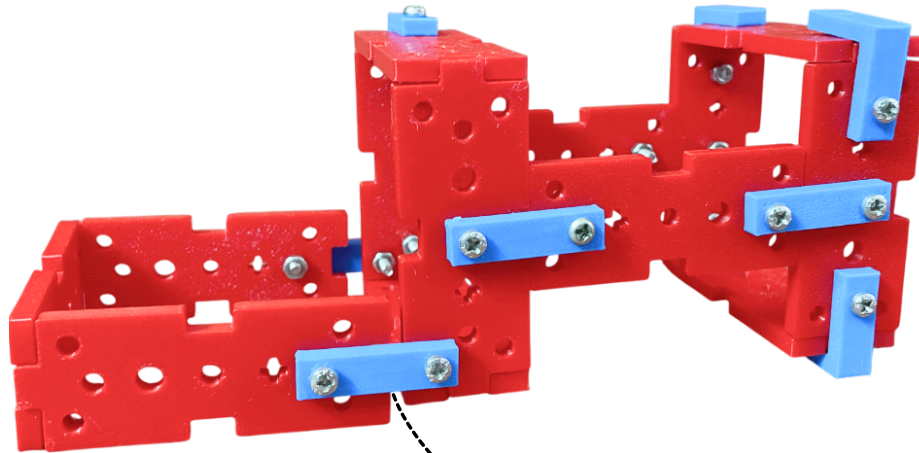


small screw - x2



nut - x2

step 6: joining the front & back chassis

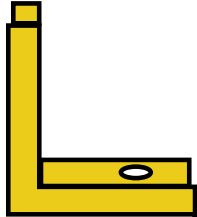


Join the front and back chassis using the I clamp as shown.

materials needed



I clamp - x2



L clamp - x2



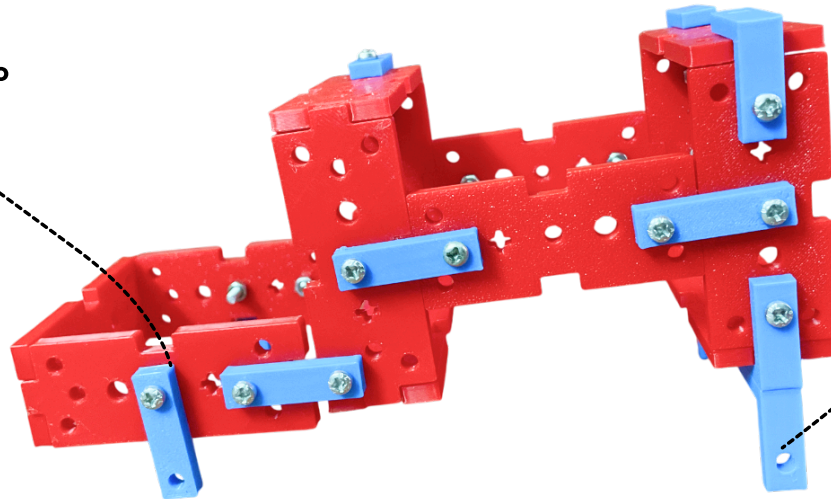
small screw - x4



nut - x4

step 7: add the wheel supports

Add **2 I clamps** to the front chassis as shown



Then add **2 L clamps** to the back chassis as shown

materials needed



wheel - x4



long screw - x4

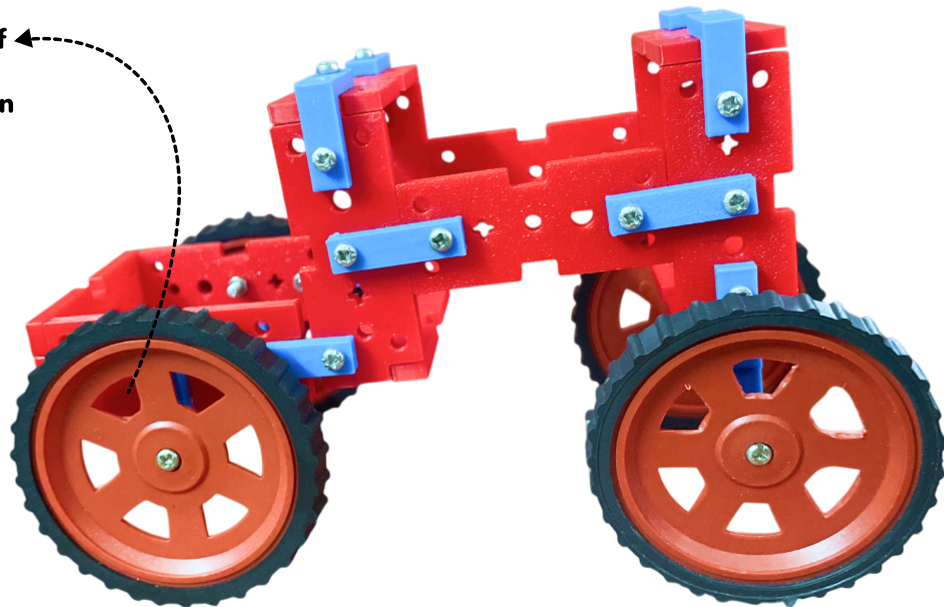


nut - x4

step 8: fix the wheel

Fix wheels to each of
the supports using
long screws as shown

Add 2 nuts to
tighten each screw.



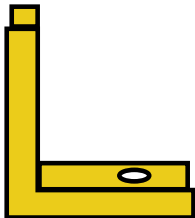
materials needed



small screw - x2

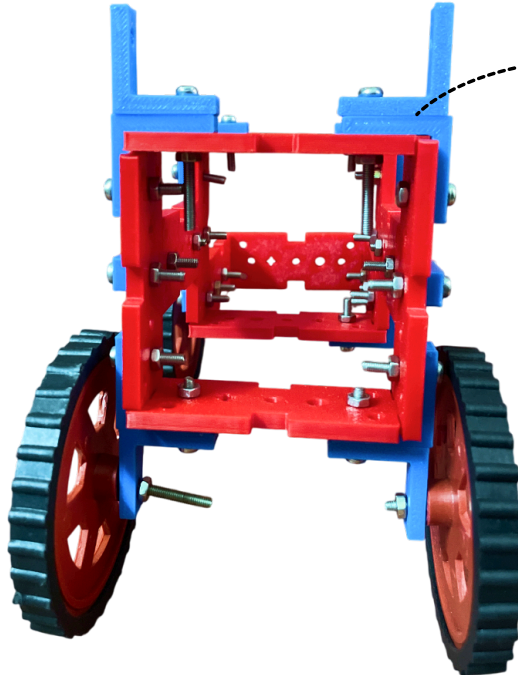


nut - x2



L clamp - x2

step 9: add the pulley supports

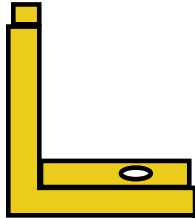


Use **2x L clamps** as the pulley supports, place them on the back of the chassis as shown.

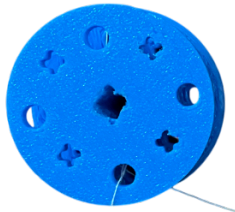
materials needed



rope - 10cm - x1

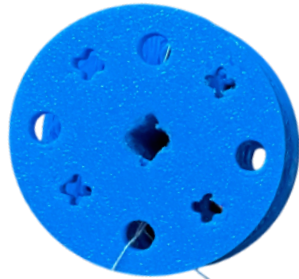


L clamp - x1



pulley - x1

step 10: add the rope to the pulley



Take a **rope of 30cm** and join one end to the pulley and the other end to the **L clamp**.

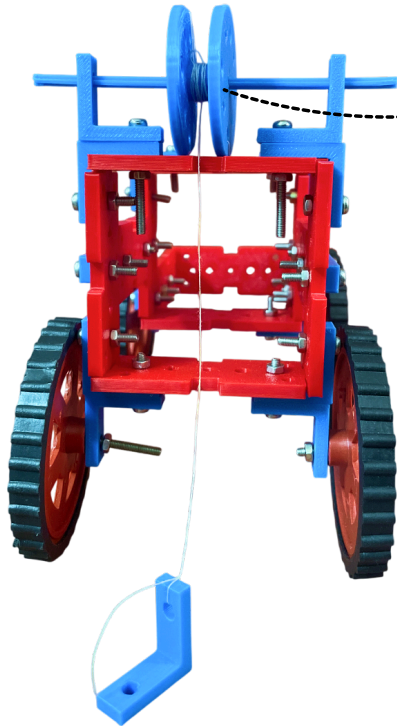


materials needed

long axle x1



step 11: complete the pulley assembly



Join the pulley to the back of the truck by placing it in between the the pulley supports and pass a **long axle** through it.

With this is
your Tow truck
is ready to tow!